

Ex 9.8 E

Q36 $\sin(\tan^{-1} x)$

① Calculate Taylor series
(find derivatives)

② Operations on power series ✓

$$\sin x = 1 - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

↑

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

replace $-x$ by x^2

$$\sin(\tan^{-1}x)$$

$$= \cancel{x} - \frac{(\tan^{-1}x)^3}{3!} + \frac{(\tan^{-1}x)^5}{5!} - \frac{(\tan^{-1}x)^7}{7!} + \dots$$

$$(\tan^{-1}x)^2$$

		x	x^3	x^5	x^7
		1	$-\frac{1}{3}$	$\frac{1}{5}$	$-\frac{1}{7}$
x	1	1	$-\frac{1}{3}$	$\frac{1}{5}$	$-\frac{1}{7}$
x^3	$-\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{9}$	$-\frac{1}{15}$	$\frac{1}{63}$
x^5	$\frac{1}{5}$	$\frac{1}{5}$	$-\frac{1}{15}$	$\frac{1}{125}$	$-\frac{1}{1575}$
x^7	$-\frac{1}{7}$	$-\frac{1}{7}$	$\frac{1}{63}$	$-\frac{1}{1575}$	$\frac{1}{11025}$

$$\therefore (\tan^{-1}x)^2 = x^2 - \frac{2}{3}x^4 + \frac{23}{45}x^6 + \dots$$

$$(\tan^{-1} x)^3 = (\tan^{-1} x)^2 (\tan^{-1} x)$$

$\underbrace{\hspace{1cm}}$
obtained just now

$(\tan^{-1} x)^2$		x^2	x^4	x^6
$\tan^{-1} x$		1	$-\frac{2}{3}$	$\frac{23}{45}$
x	1	1	$-\frac{2}{3}$	$\frac{23}{45}$
x^3	$-\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{9}$	
x^5	$\frac{1}{5}$	$\frac{1}{5}$		
x^7	$-\frac{1}{7}$			

$$\therefore (\tan^{-1} x)^3 = x^3 - x^5 + \frac{14}{15} x^7 + \dots$$

$$(\tan^{-1} x)^5 = \underbrace{(\tan^{-1} x)^3 (\tan^{-1} x)^2}_{\text{obtained just now}}$$

$(\tan^{-1} x)^2$		x^2	x^4	x^6
$(\tan^{-1} x)^3$		1	$-\frac{2}{3}$	$\frac{23}{45}$
x^3	1	1	$-\frac{2}{3}$	.
x^5	-1	-1	.	.
x^7	$\frac{14}{15}$	.	.	.

$$\therefore (\tan^{-1} x)^5 = x^5 - \frac{5}{3} x^7 + \dots$$

$$(\tan^{-1} x)^7 = x^7 + \dots$$

$$\sin(\tan^{-1} x) = ?$$

	x	x^3	x^5	x^7		x	x^3	x^5	x^7
$\tan^{-1} x$	1	$-\frac{1}{3}$	$\frac{1}{5}$	$-\frac{1}{7}$		1	$-\frac{1}{3}$	$\frac{1}{5}$	$-\frac{1}{7}$
$(\tan^{-1} x)^3$	0	1	-1	$\frac{14}{15}$	$\times \frac{1}{3!}$	0	$-\frac{1}{6}$	$\frac{1}{6}$	$-\frac{7}{45}$
$(\tan^{-1} x)^5$	0	0	1	$-\frac{5}{3}$	$\times \frac{1}{5!}$	0	0	$\frac{1}{120}$	$-\frac{1}{72}$
$(\tan^{-1} x)^7$	0	0	0	1	$\times \frac{1}{7!}$	0	0	0	$-\frac{1}{5040}$

$$\therefore \sin(\tan^{-1} x) = x + \left(-\frac{1}{3} - \frac{1}{6}\right)x^3 + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{120}\right)x^5 + \left(-\frac{1}{7} - \frac{7}{45} - \frac{1}{72} - \frac{1}{5040}\right)x^7 + \dots$$